

Name: _____

Double Award Science – Applied Unit 2 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision. **You will also need the resource pack from the correct year.**

Year	Score /75	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	16	18	22	26	30	34	40
2018	17	20	25	30	35	40	45
2019	13	16	22	28	35	42	49
2022	6	8	16	25	35	42	50
2023	5	9	17	25	33	41	49

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE – NEW**

3445UB0-1

**APPLIED SCIENCE (Double Award)****Unit 2: Space, Health and Life****HIGHER TIER**

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 30 minutes

	For Examiner’s use only		
	Question	Maximum Mark	Mark Awarded
Section A	1. (a)(b)(c)(d)	19	
	1. (e)	6	
Section B	2.	14	
	3.	10	
	4.	10	
	5.	10	
	6.	6	
	Total	75	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 1(e) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

Section A

*Answer **all** questions in the spaces provided.*

Use the information in the separate Resource Folder to answer the following questions.

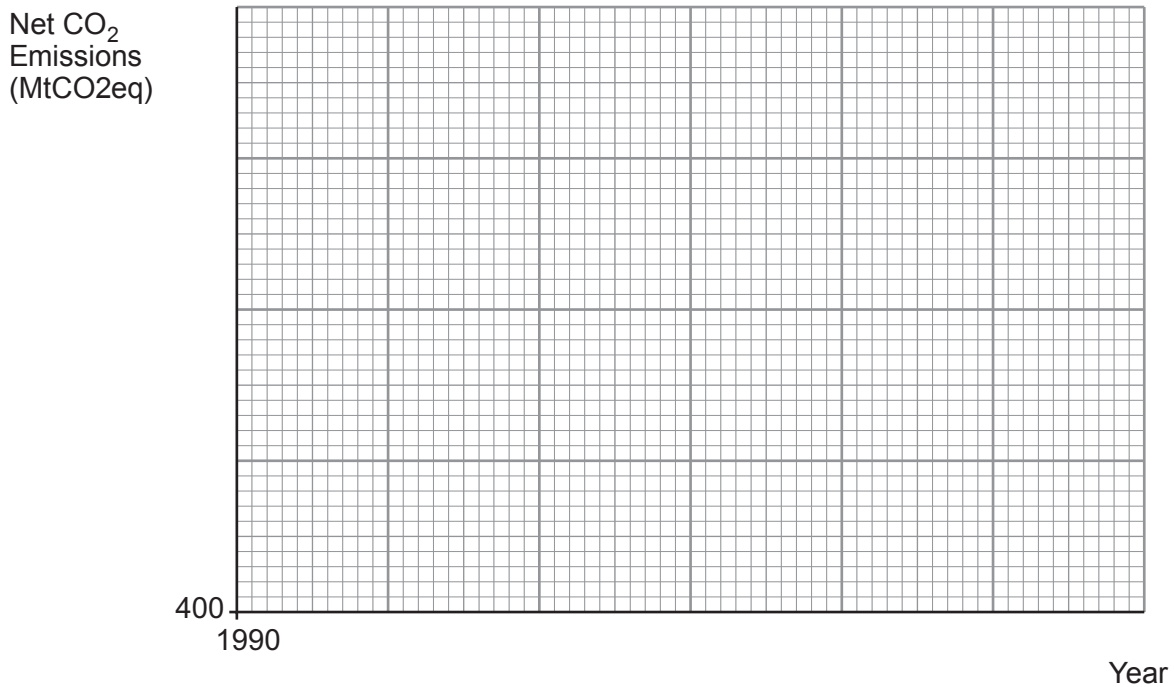
1. (a) (i) Use the information in **Table 1** to describe the trend in emissions of HFC between 1990 and 2014. [2]

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- (ii) Use the information in **Table 1** to plot a graph of net CO₂ emissions from the UK between 1990 and 2014. [4]



- (iii) Describe the trend shown in the graph. [2]

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- (iv) Describe how your graph (opposite) compares with the information in **Graph 1**. [2]

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- (v) Use your knowledge of the carbon cycle to account for the seasonal fluctuations in CO₂ concentration shown in **Graph 1**. Give reasons for your answer. [3]

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- (b) Use the information in **Tables 2** and **3** and the equation:

$$\text{greenhouse contribution} = (\text{concentration in the atmosphere}) \times (\text{multiplier})$$

to complete the table below.

[3]

gas	greenhouse contribution					
	multiplier	pre-industrial baseline	natural additions	man-made additions	total relative contribution	% of total contribution
carbon dioxide	1	288 000		11 880	368 400	72.37
methane	21	17 808	12 117		36 645	

- (c) Use the information in **Tables 2** and **4** to describe the effect on the data in the last column of **Table 2** when water vapour is included. [2]

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- (d) State **one** conclusion that can be made from the information shown in **Chart 1** on page 4 of the resource folder. [1]

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(e) Explain whether **Graph 2** and **Graph 3** support the view that global warming is due to solar activity and not CO₂ concentration. [6 QER]

Examiner
only

[illegible]

Section B

Answer all the questions in the spaces provided.

2. A class of students were using dice to model radioactive decay. There were 10 groups of students in the class. Each group of students had 90 dice.



- (a) (i) Describe how they should carry out the experiment.

[3]

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- (ii) State the number of dice you would expect to land with a 6 facing up on the first throw in **each** group.

[1]

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- (iii) Each group's results were added together to give the class results. Give **one** advantage of using a bigger sample size.

[1]

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.....

(b) The table shows the results for the **whole class**.

Throw	Number of sixes	Number of dice remaining
0	0	900
1	146	754
2	124	630
3	103	527
4		442
5	72	370
6	61	309
7	52	257
8	38	219
9	34	

(i) **Complete** the table.

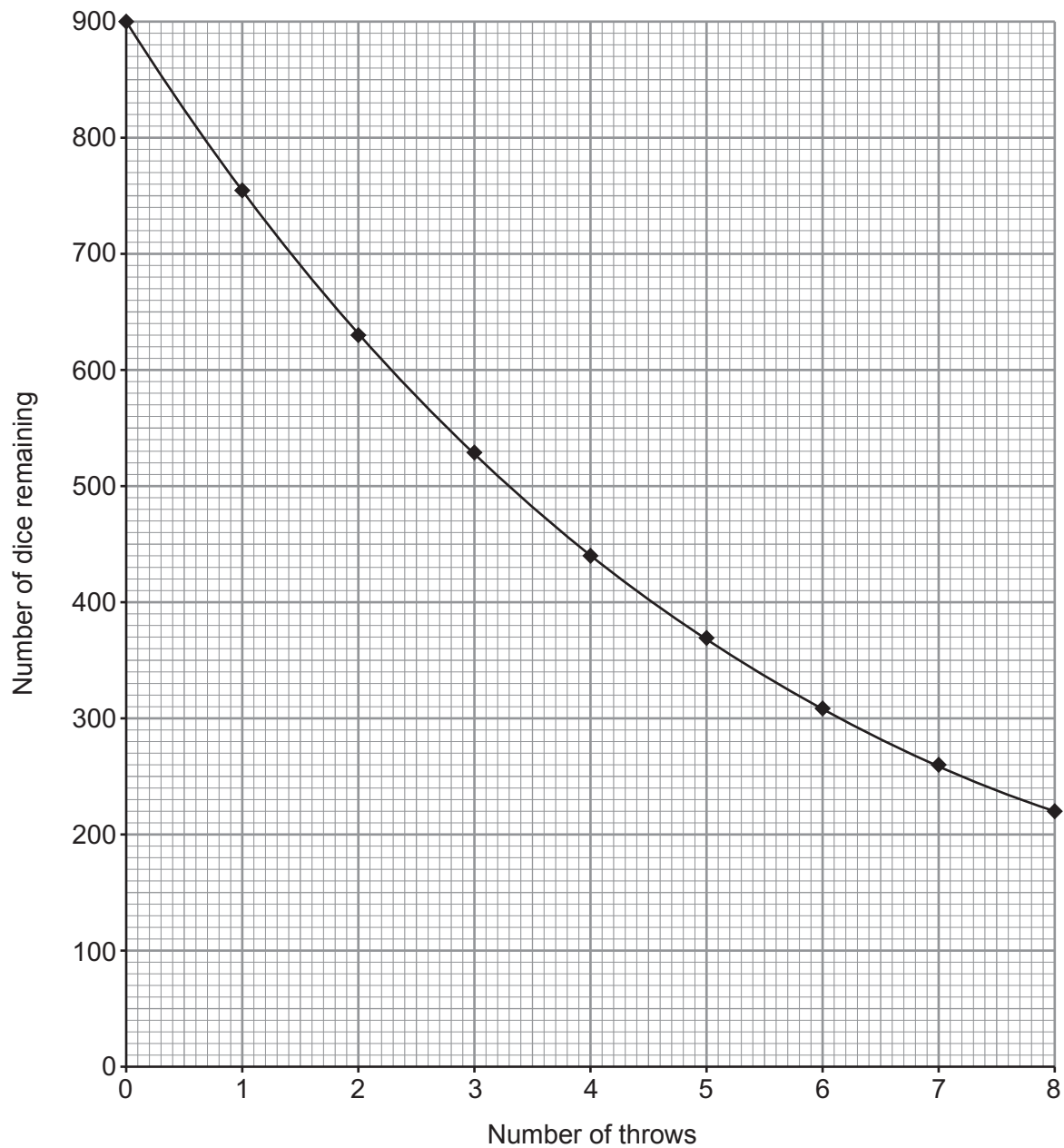
[2]

(ii) Use the data in the table to estimate the half-life of the 'dice'.

[1]

half-life = throws

(c) Some of the data in the table is plotted on the grid below.



(i) **Draw lines** on your graph to find the half-life of the dice.

[2]

half-life = throws

(ii) State **one** reason why finding the half-life using the graph is a better method than the one used in part (b)(ii).

[1]

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- (d) Radioisotopes have many medical uses. One use is to sterilise blood products for transfusion.



- (i) Select from the table below the most suitable radioisotope for this purpose. [1]

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Radioisotope	Half-life	Type of emissions
bismuth-213	46 minutes	alpha
chromium-51	28 days	gamma
caesium-137	30 years	beta and gamma
bismuth-209	2×10^{19} years	alpha

- (ii) Give **two** reasons for your choice. [2]

1.

2.

3. Huntington's disease (HD) is an inherited disease that causes nerve cells in the brain to degenerate. People are born with the defective gene, but symptoms usually don't appear until middle age. Early symptoms of HD may include uncontrolled movements, clumsiness or balance problems. The disease is caused by an inherited defect in a single gene. With only one copy of the defective dominant allele (**H**), a person can develop the disease.

(a) State the meaning of the following terms:

[3]

(i) allele

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(ii) dominant allele

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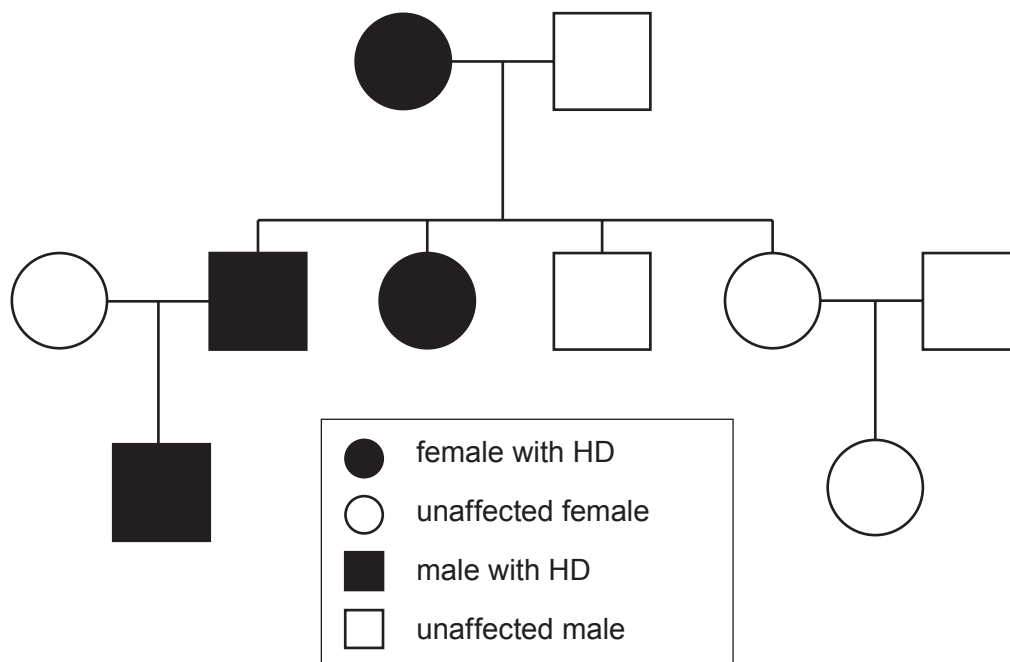
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(iii) genotype

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- (b) The diagram shows a family tree identifying those who are unaffected (do not have HD) and those with HD.



Draw **two** Punnett squares and **compare** the percentage chances of a child developing HD if:

- one parent is unaffected and one parent is heterozygous
- both parents are heterozygous

[5]

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(c) Discuss the ethical problems which may arise when an individual undergoes genetic screening and finds out they have the allele for HD. [2]

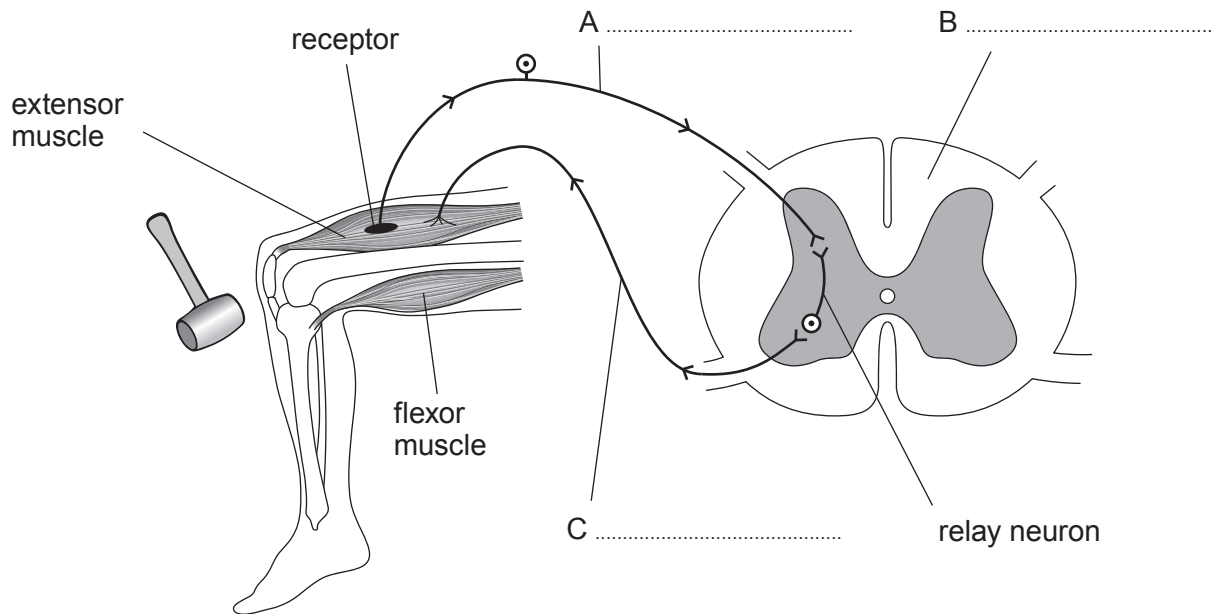
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4. The diagram shows a reflex arc.

Examiner
only



(i) **Label** the diagram. [3]

(ii) A doctor checks the reflexes of a patient by tapping the knee sharply which should result in the lower leg moving up quickly. Describe how the tap on the knee results in the lower leg moving up. Refer to the information in the diagram in your answer. [4]

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(iii) Describe the function of the antagonistic pair of muscles shown in the diagram. [3]

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5. Since October 2011 there has been a minimum charge of 5p for carrier bags in Wales.

- (a) (i) Explain in terms of our natural resources why we should reduce our use of plastic carrier bags. [2]

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- (ii) Describe **three** benefits to the environment if fewer plastic bags are used. [3]

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- (b) Before the introduction of charges, 350 million plastic bags were used every year. Five years later this figure had dropped to 70 million used every year. Before a charge was introduced in England, 7 500 million plastic carrier bags were used per year. If the same success is achieved as in Wales, calculate the number of plastic carrier bags that will be in use per year in England five years later. [3]

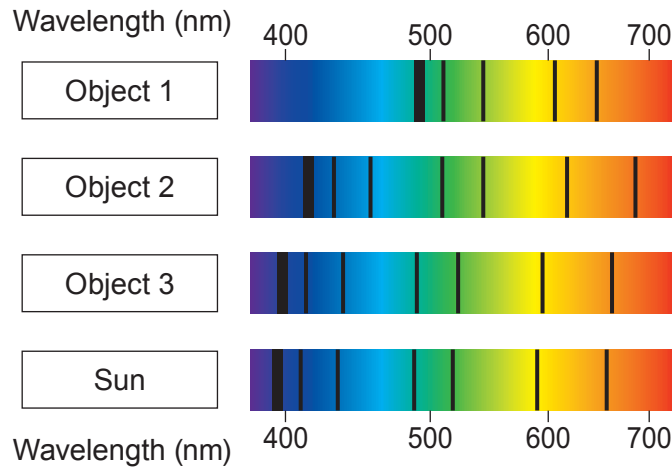
number =

- (c) When researchers collected reused plastic bags from supermarket shoppers they found large numbers of bacteria in almost all carrier bags. Explain why this is a concern. [2]

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6. The diagram below shows absorption spectra from four different objects in space. The bottom spectrum is from our Sun.



- (a) Explain how an absorption spectrum is produced.

[3]

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- (b) Explain how you can deduce, from the spectra, that objects 1, 2 and 3 have the same chemical composition as the Sun and are moving away from Earth at different speeds.

[3]

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END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1

H

Hydrogen

1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

65 Zn Zinc 30	63.5 Cu Copper 29	59 Ni Nickel 28	59 Co Cobalt 27	56 Fe Iron 26	55 Mn Manganese 25	52 Cr Chromium 24	51 V Vanadium 23	48 Ti Titanium 22	45 Sc Scandium 21
112 Cd Cadmium 48	108 Ag Silver 47	106 Pd Palladium 46	103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	96 Mo Molybdenum 42	93 Nb Niobium 41	91 Zr Zirconium 40	89 Y Yttrium 39
201 Hg Mercury 80	197 Au Gold 79	195 Pt Platinum 78	192 Ir Iridium 77	190 Os Osmium 76	186 Re Rhenium 75	184 W Tungsten 74	181 Ta Tantalum 73	179 Hf Hafnium 72	139 La Lanthanum 57

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0


GCSE – NEW
3445UB0-1


S18-3445UB0-1

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life
HIGHER TIER
MONDAY, 11 JUNE 2018 – MORNING
1 hour 30 minutes

Section A	For Examiner’s use only			
	Question	Maximum Mark	Mark Awarded	
	1.(a)(b)(c)	19		
	1.(d)	6		
	Section B	2.	16	
		3.	12	
		4.	12	
		5.	10	
		Total	75	

ADDITIONAL MATERIALS

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INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

Section A

Answer **all** questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

1. (a) Use the information in **Table 1** to answer the following questions.
- (i) Europa is an asteroid. Estimate its temperature and orbital period.

[2]

Temperature = °C

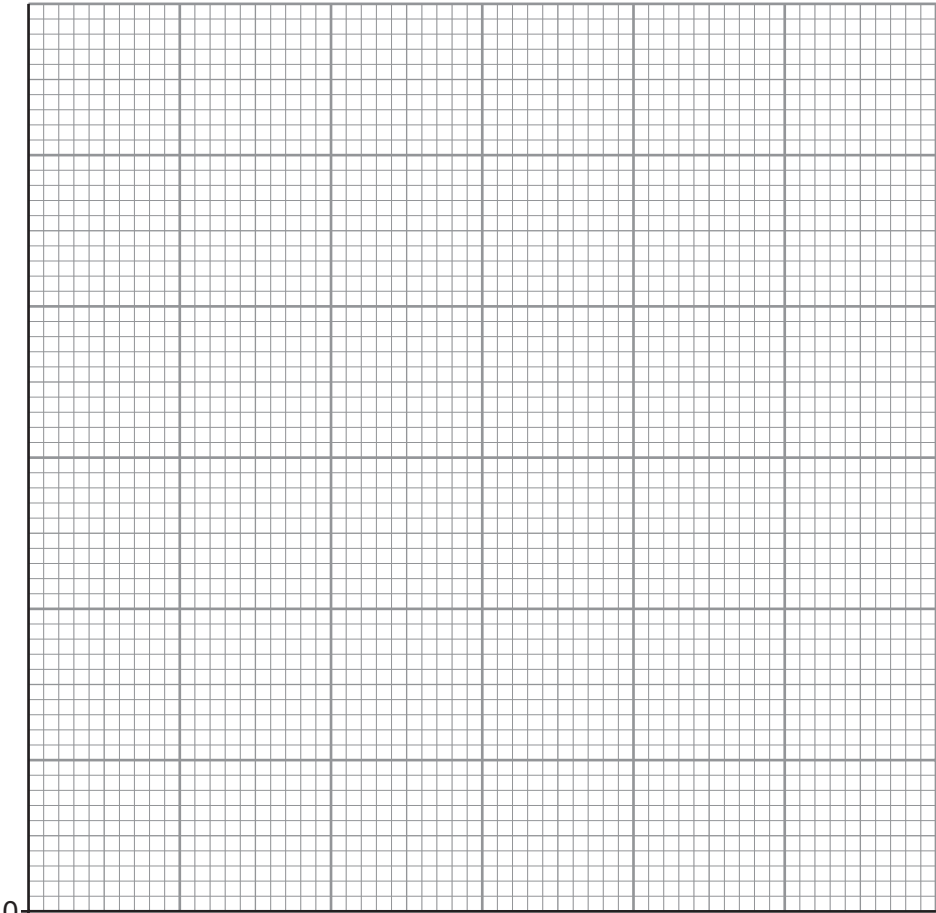
Orbital period = days
- (ii) Pluto is no longer classed as a planet. State **one** reason why.

[1]

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- (iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line.

[4]

Orbital
Velocity
(km/s)



Distance from Sun (AU)

- (iv) Orbital velocity is not proportional to the distance from the Sun.
Explain how your graph shows this to be true. [2]

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- (b) Use your knowledge and the information on **pages 4 and 5** to answer the following questions.

- (i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

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- (ii) State **two** ways in which human activities may be affected by solar flares. [2]

1.

2.

- (iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

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- (c) Use the information in **Diagram 1** to describe the differences between Aristotle's model of the Solar System and the 2006 model. [4]

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(d) Use the information in **Diagram 3** to answer the following questions.

(i) Explain what you can deduce about the motion of the three galaxies. [3]

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(ii) Explain what you can deduce about the chemical composition of the three galaxies. [3]

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Section B

2. Velothon Wales is a weekend of road cycling events. The Bloodwise cycling team raises funds to help beat blood cancer.

(a) Three cyclists are monitored before the Velothon. They have blood samples taken.

The results of their blood tests are shown in the table below. Normal ranges are also given.

Name of cyclist	Red blood cell count ($10^6/\text{cm}^3$)	White blood cell count ($10^3/\text{cm}^3$)	Platelet count ($10^3/\text{cm}^3$)
<i>Normal range</i>	<i>4.4 - 5.8</i>	<i>3.9 - 10.8</i>	<i>130 - 400</i>
Brian	5.5	35.1	240
Gareth	2.2	6.6	115
Jonathan	4.7	8.8	320

- (i) Suggest which cyclist needs treatment for an infection. [2]

Cyclist

Reason

- (ii) Suggest which cyclist needs treatment to raise their haemoglobin level. [2]

Cyclist

Reason

- (iii) Suggest which cyclist will experience a problem if he has a fall and cuts himself. [2]

Cyclist

Reason

- (b) Explain how the structure of veins and capillaries are related to their function. [4]

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- (c) (i) During a training ride, Jonathan accelerated from rest over a distance of 10 m in 2.5 s then travelled another 96 m at a constant speed in 6 s.
Use the equations:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

and

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration of the cyclist.

[4]

$$\text{Acceleration} = \dots\dots\dots \text{ m/s}^2$$

- (ii) Jonathan has a height of 1.9 m. He is advised his maximum BMI should be 22.
Use the equation:

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

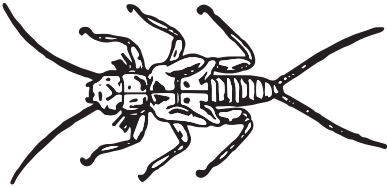
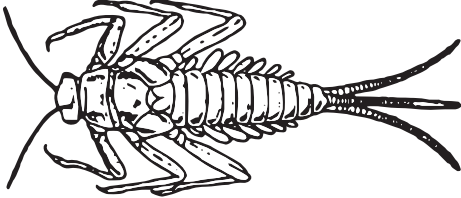
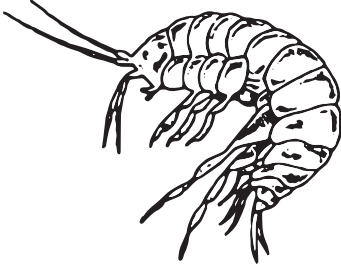
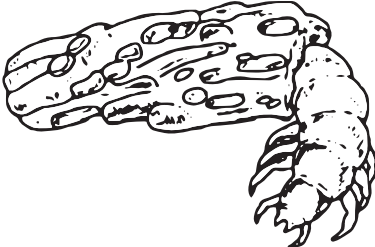
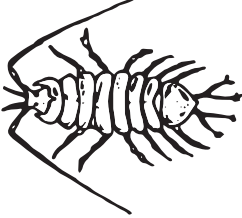
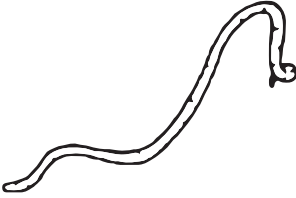
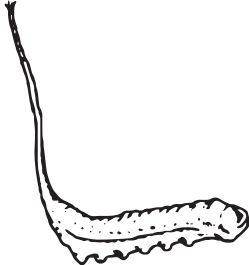
to calculate his maximum mass.

[2]

$$\text{Mass} = \dots\dots\dots \text{ Kg}$$

Turn over.

3. Water pollution can be monitored using invertebrates as indicators. Some indicator species are shown below.

Water quality	Indicator species	
Clean water	 Stonefly nymph	 Mayfly nymph
Some pollution	 Freshwater shrimp	 Caddisfly larva
Moderate pollution	 Water louse	 Bloodworm
High pollution	 Sludgeworm	 Rat-tailed maggot
Very high pollution	No life	

(Not drawn to scale.)

(a) Describe how you would carry out an investigation to determine water quality in a stream using invertebrates as indicator species. [6 QER]

Examiner
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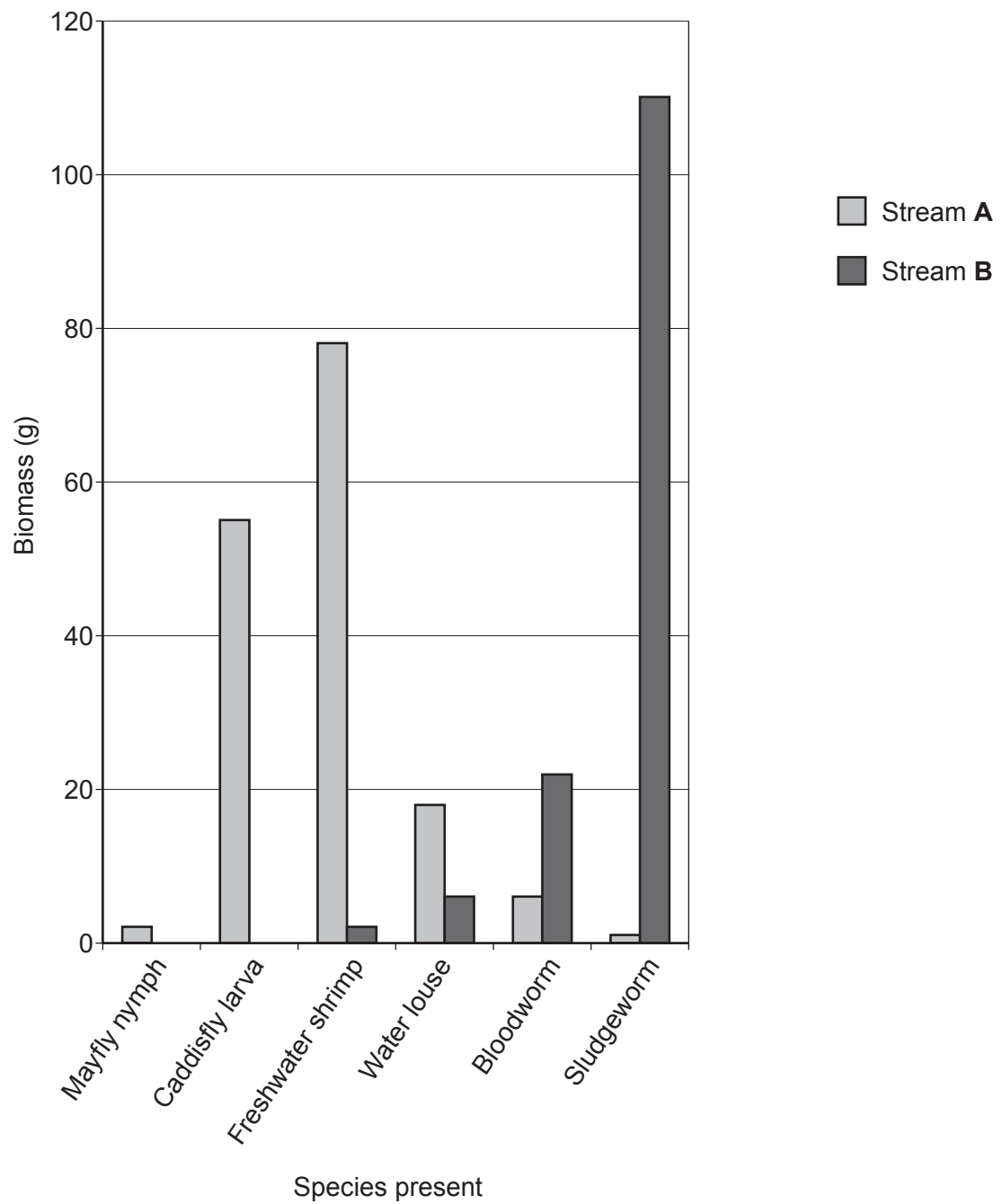
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(b) The results from such an investigation are shown in the chart below.



- (i) Explain what conclusions can be made about the water quality in stream **A** and stream **B**. [4]

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- (ii) Explain why it is difficult to obtain repeatable data in this experiment. [2]

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Examiner
only

12

4. (a) Green plants capture a small percentage of solar energy during the process of photosynthesis. The table below shows how the absorption of light by a green plant depends on the wavelength of the visible light.

Wavelength of visible light (nm)	Absorption (%)
400	85
450	100
500	2
550	12
600	18
650	35
700	75

- (i) Describe how the absorption of light depends on the wavelength of visible light. [3]

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- (ii) Electromagnetic waves travel at 3×10^8 m/s in space.

Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the frequency of light which is most strongly absorbed. [3]

$$(1 \text{ nm} = 1 \times 10^{-9} \text{ m})$$

Frequency = Hz

Examiner
only

- (b) The table shows what happens to the energy taken in each day by organisms in the food chain below.

grass → grasshopper → shrew → fox

Organisms	Energy per day (MJ)		
	As waste	Released during respiration	Used for growth
grass	6	10	8
grasshopper	14	22	3
shrew		26	4
fox	24	32	4

- (i) The shrew releases 60% of its energy during respiration and for growth. Calculate how much energy is released by the shrew as waste.

[2]

Energy as waste = MJ

- (ii) Explain why the amount of energy released during respiration by each organism increases through the food chain.

[2]

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- (iii) Sometimes plant seeds will stick to the coat of a fox. Explain how this benefits the plant.

[2]

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5. Nuclear medicine involves the use of radioactive substances in the diagnosis and treatment of disease. Nuclear medicine records radiation emitted from within the body rather than radiation that is generated by external sources like X-rays. Nuclear medicine scans differ from radiology as the emphasis is not on imaging but on organ function.

Information about some isotopes that emit gamma (γ) rays is given in the table.

Radioisotope	Half-life	Energy of γ rays
caesium-137	30.17 years	0.662 MeV
cobalt-60	5.26 years	1.17 MeV
iodine-125	59.6 days	31.4 keV
thallium-201	73.0 hours	71.0 keV
palladium-103	17.0 days	21.0 keV

- (a) (i) Describe the nature of a gamma (γ) ray.

[2]

- (ii) Explain why radioisotopes that emit gamma rays are suitable for examining organ function when injected into the body.

[2]

- (iii) Explain which is the most suitable radioisotope from the table to inject into the human body as part of a gamma camera investigation.

[2]

- (b) In order to treat prostate cancer, palladium-103 pellets are placed directly into the prostate gland. They remain permanently in place. The gamma emissions from the pellets are almost undetectable when their activity drops to $\frac{1}{32}$ of its original value. Calculate the time taken for this reduction to occur. [4]

Time = days

END OF PAPER

Examiner
only

10

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2																									
7 Li Lithium 3		9 Be Beryllium 4		11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10																					
23 Na Sodium 11		24 Mg Magnesium 12		27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18																					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

Key

relative atomic mass

Ar

Symbol

Name

Z

atomic number



Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3445UB0-1



S19-3445UB0-1

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life

HIGHER TIER

FRIDAY, 7 JUNE 2019 – AFTERNOON

1 hour 30 minutes

	For Examiner's use only		
	Question	Maximum Mark	Mark Awarded
Section A	1.	19	
	2.	6	
Section B	3.	8	
	4.	16	
	5.	8	
	6.	11	
	7.	7	
	Total	75	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

- Use black ink or black ball-point pen.
- Write your name, centre number and candidate number in the spaces at the top of this page.
- Answer **all** questions.
- Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets at the end of each question or part-question.
- Question **7(b)** is a quality of extended response (QER) question where your writing skills will be assessed.
- You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
- A periodic table is printed on page 20.

Section A

Examiner
only

Answer **all** questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

1. (a) Use the information in **Figure 3** to answer the following questions.
- (i) A man drinks six 175 ml glasses of wine, three pints of lager and two 25 ml glasses of spirits. Calculate the total number of units of alcohol consumed. [1]

Number of units =

- (ii) He stopped drinking at 11 pm. He is planning to drive at 1 pm the following day. Explain whether he can be **certain** that all units of alcohol will have been processed by this time so it will be safe for him to drive. Show your calculations. [2]

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- (b) Use the information in **Figures 4 and 5** to answer the following questions.
- (i) Explain why it is difficult to accurately compare the nutrition facts for the three cereals shown. [2]

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- (ii) Explain which cereal you would recommend for a 7-year-old child. [2]

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- (c) Calculate the daily personal energy requirement (PER) for a 70 kg athlete who trains for 4 hours a day. [3]

PER = units

(d) Use the information in **Figures 6, 7 and 8** to answer the following questions.

- (i) State the mass at which a person of height 1.7 m has a high risk of developing health problems. [1]

Mass = kg

- (ii) Explain whether you agree with the statement that 'BMI is proportional to body mass'. [2]

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- (iii) Calculate the BMI for a person of body mass 75 kg and height 1.6 m. [2]

BMI = kg/m²

- (e) (i) A woman suffers from maturity onset diabetes of the young (MODY). This is caused when one parent is heterozygous for the condition. Her partner is not diabetic. Complete the Punnett square, using **D** as the dominant allele, to show the possible genotypes of the children. [3]

- (ii) State the chance of a child being born with MODY. [1]

Chance =

Examiner
only

2. (a) Use the information in **Figure 1** to explain how a person's cardiovascular system improves in the first year after they stop smoking. [3]

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- (b) Use the information in **Figure 2** to explain whether the percentage of smokers who have quit has consistently risen in the 20-year period ending 2016. [3]

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6

Section B

Answer all questions in the spaces provided.

Examiner
only

3. The abundance of a species in a habitat can be investigated by using quadrats for random sampling.

(a) Describe how you would use a quadrat to investigate the population of snails in a habitat. [3]

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(b) The results of an investigation into the variety of snails found in two habitats are shown below. Each quadrat measured 1 m by 1 m. The area of habitat X was 12 m² and the area of habitat Y was 15 m².

Habitat X

Quadrat	Number of snails				Total number of snails
	Species A	Species B	Species C	Species D	
1	0	2	3	1	6
2	0	0	3	2	5
3	0	1	3	1	5

Habitat Y

Quadrat	Number of snails				Total number of snails
	Species A	Species B	Species C	Species D	
1	0	0	1	2	3
2	0	0	4	3	7
3	0	0	1	3	4

(i) Use the data to compare the distribution of snail species in the two habitats. [4]

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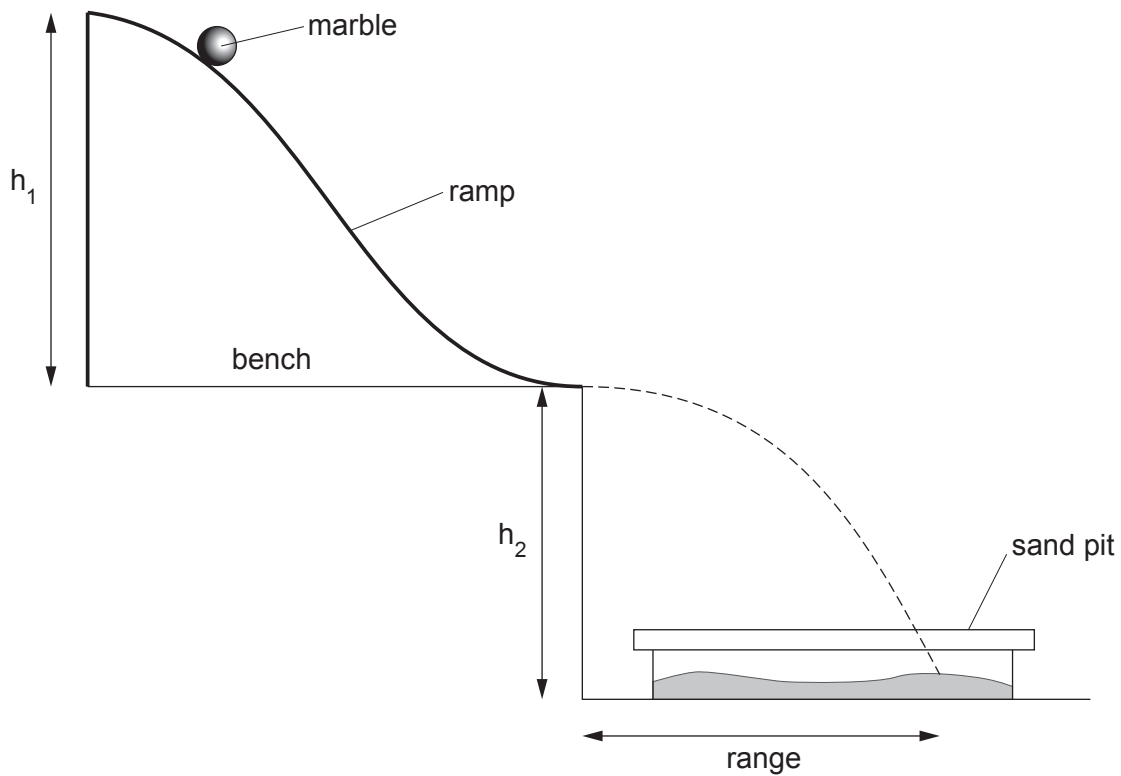
(ii) One student claims the investigation is not valid as the area of the habitats was not the same. Explain whether you agree with the claim. [1]

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4. A model ski jump can be set up on a bench as shown in the diagram below.



The marble is released from rest at the top of a ramp. It leaves the bottom of the ramp with a certain velocity. It moves through the air and lands in a sand pit. The horizontal distance travelled in the air is called the **range**.

The velocity of the marble as it leaves the ramp is calculated using the equation:

$$\text{velocity} = \frac{\text{range}}{\text{time to travel the range}}$$

The acceleration of the marble down the ramp is calculated using the equation:

$$\text{acceleration} = \frac{\text{velocity of marble as it leaves the ramp}}{\text{time to travel down the ramp}}$$

- (a) The following measurements were recorded during one trial of the experiment.

Range = 45 cm

Time to travel the range = 0.5 s

Time to travel down the ramp = 1.2 s

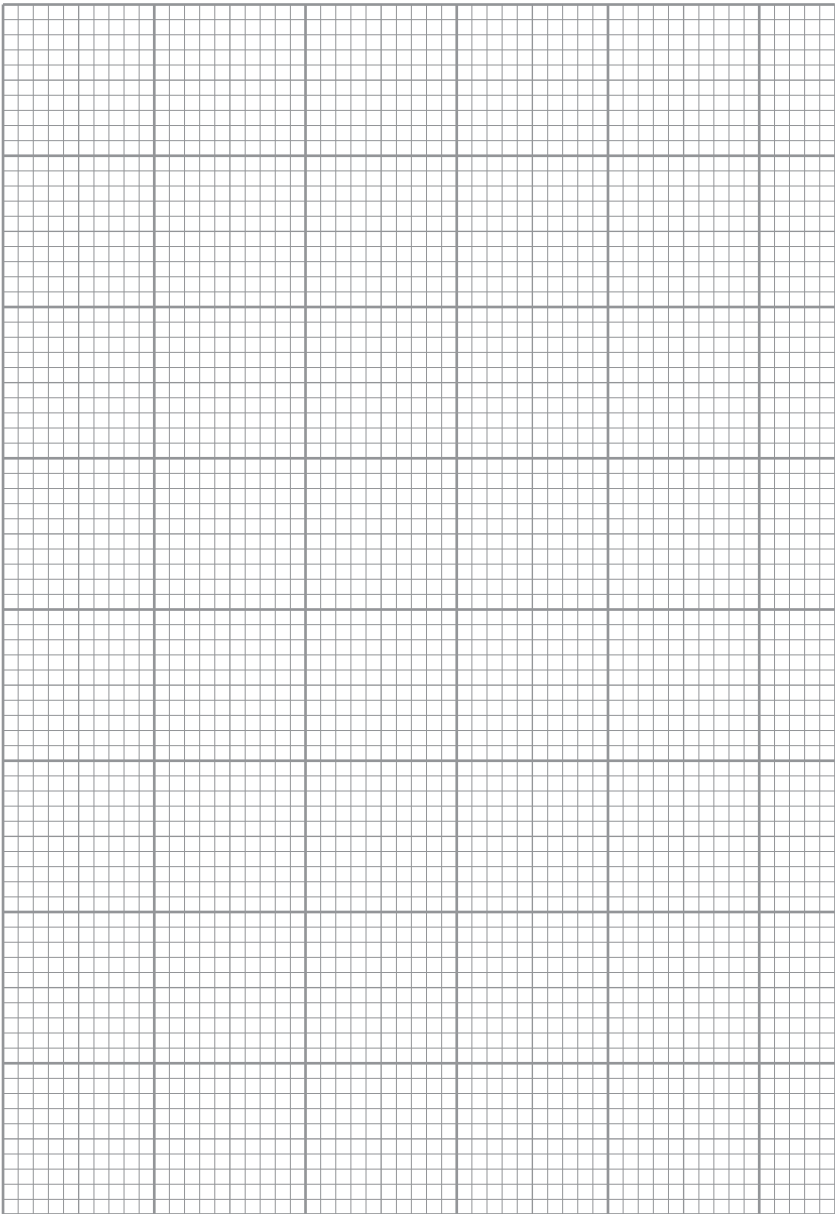
Use the equations and information above to calculate the acceleration of the marble down the ramp in m/s^2 . [4]

Acceleration = m/s^2

- (b) An experiment is then carried out to find out how the acceleration of the marble depends on the height h_1 . Throughout this experiment height h_2 is constant. The results of the experiment are shown in the table below.

Height h_1 (m)	Acceleration (m/s^2)
0	0
0.2	0.14
0.5	0.36
0.6	0.43
0.9	0.62
1.0	0.74

- (i) Use the data in the table to plot a graph on the grid below and draw a suitable line. [3]



- (ii) Use the graph to find the height required to produce an acceleration of 0.5 m/s^2 . [1]

Height =

- (iii) Describe how the experiment can be extended to determine how the range depends on the height h_2 . [2]

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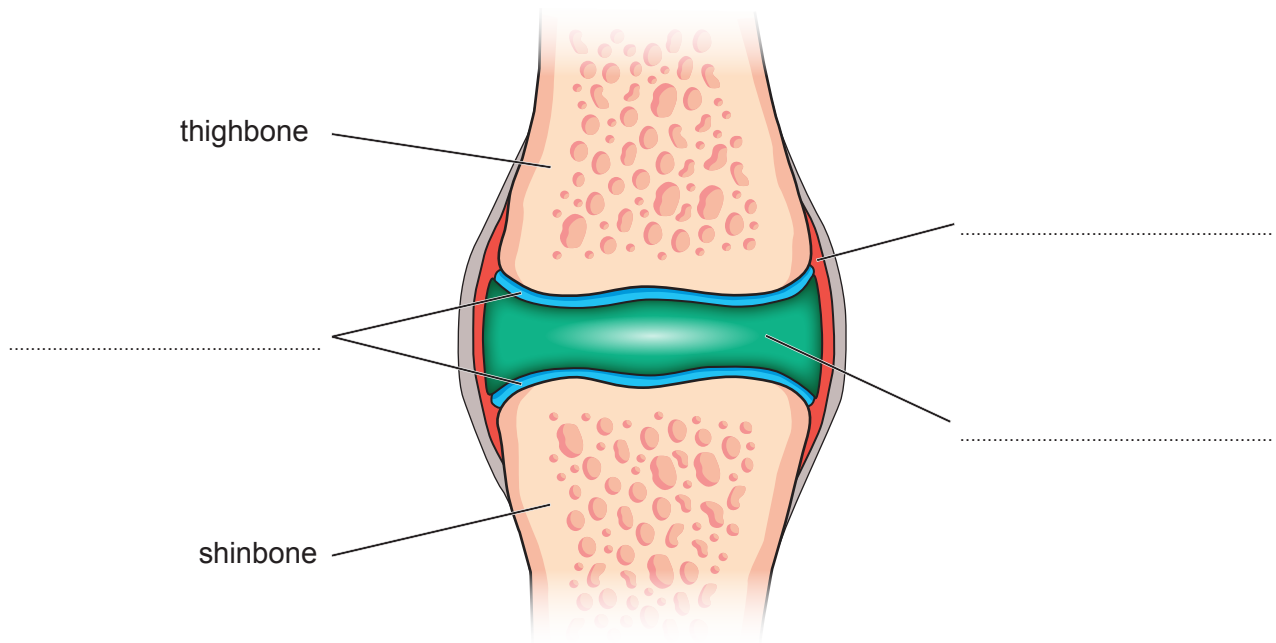
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- (c) Ski jumpers place their knee joints under considerable strain during their career. The knee is easily injured.

A simplified diagram of a knee joint is shown below.



- (i) **Complete** the labelling of the diagram above. [3]

- (ii) Explain why a ski jumper will not be able to compete if they damage the cartilage in their knee joints. [2]

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- (iii) An X-ray was taken of a ski jumper's leg following an accident.



State the name of this **type** of fracture.

[1]

16

Examiner
only

5. In normal circumstances, nitrogen in the soil is recycled and causes no harm to aquatic organisms in nearby streams or rivers.

(a) Describe how nitrogen is recycled in the soil. [4]

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(b) Use of artificial fertilisers increases the level of nitrogen in the soil. Explain why this becomes harmful to aquatic organisms. [4]

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8

6. Humans are protected against disease by vaccination programmes and the use of antibiotics.

(a) Measles is a highly infectious viral illness that is uncommon in the UK because of the effectiveness of vaccination. The measles vaccine provides life-long protection against the disease.

(i) Explain how vaccines work. [2]

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(ii) A flu vaccine is also available. Unlike the measles vaccine, it needs to be given every year and is not 100% effective. Explain why a flu vaccine does not provide life-long protection. [4]

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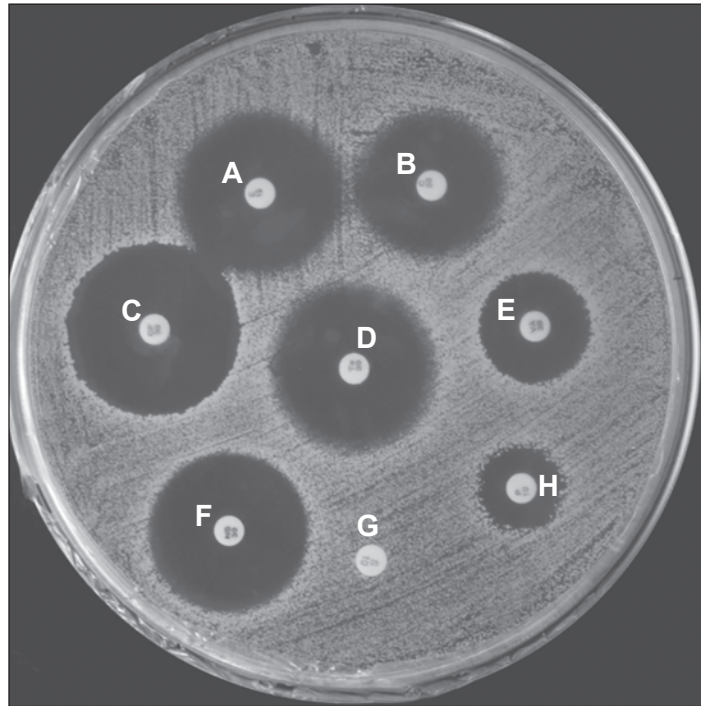
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- (b) Doctors prescribe patients antibiotics to help fight bacterial infections. There are a variety of antibiotics available.

The photograph below shows the results of an experiment to determine the effectiveness of different antibiotics against one type of bacterium.

Eight different antibiotics were tested. Each disc contains one of the antibiotics (**A** to **H**).



- (i) Evaluate the effectiveness of the antibiotics **A** to **H**.

[3]

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- (ii) A disc containing sterile water could be included to act as a control (not shown in the photograph). State the purpose of this control disc.

[2]

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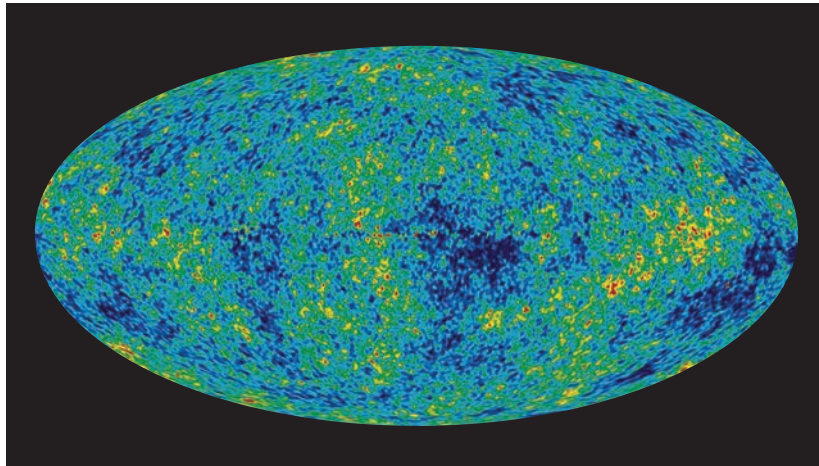
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7. In 1802, spectra from stars and galaxies were first observed.

In the 1940s the Steady State Theory of the origin of the Universe was proposed by Sir Fred Hoyle and other scientists.

In 1964, American astronomers discovered there was electromagnetic interference coming from all parts of the sky at all times. This is shown in the image below.



- (a) State the name of the region of the electromagnetic spectrum that is recorded on the image shown above. [1]

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- (b) Explain the evidence that led to an alternative, different theory of the origin of the Universe to the Steady State Theory. [6 QER]

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Examiner
only

7

END OF PAPER

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																4 He Helium 2																																					
7 Li Lithium 3			9 Be Beryllium 4			11 B Boron 5			12 C Carbon 6			14 N Nitrogen 7			16 O Oxygen 8			19 F Fluorine 9			20 Ne Neon 10																																
23 Na Sodium 11			24 Mg Magnesium 12			27 Al Aluminium 13			28 Si Silicon 14			31 P Phosphorus 15			32 S Sulfur 16			35.5 Cl Chlorine 17			40 Ar Argon 18																																
39 K Potassium 19			40 Ca Calcium 20			45 Sc Scandium 21			48 Ti Titanium 22			51 V Vanadium 23			52 Cr Chromium 24			55 Mn Manganese 25			56 Fe Iron 26			59 Co Cobalt 27			59 Ni Nickel 28			63.5 Cu Copper 29			65 Zn Zinc 30			70 Ga Gallium 31			73 Ge Germanium 32			75 As Arsenic 33			79 Se Selenium 34			80 Br Bromine 35			84 Kr Krypton 36		
86 Rb Rubidium 37			88 Sr Strontium 38			89 Y Yttrium 39			91 Zr Zirconium 40			93 Nb Niobium 41			96 Mo Molybdenum 42			99 Tc Technetium 43			101 Ru Ruthenium 44			103 Rh Rhodium 45			106 Pd Palladium 46			108 Ag Silver 47			112 Cd Cadmium 48			115 In Indium 49			119 Sn Tin 50			122 Sb Antimony 51			128 Te Tellurium 52			127 I Iodine 53			131 Xe Xenon 54		
133 Cs Caesium 55			137 Ba Barium 56			139 La Lanthanum 57			179 Hf Hafnium 72			181 Ta Tantalum 73			184 W Tungsten 74			186 Re Rhenium 75			190 Os Osmium 76			192 Ir Iridium 77			195 Pt Platinum 78			197 Au Gold 79			201 Hg Mercury 80			204 Tl Thallium 81			207 Pb Lead 82			209 Bi Bismuth 83			210 Po Polonium 84			210 At Astatine 85			222 Rn Radon 86		
223 Fr Francium 87			226 Ra Radium 88			227 Ac Actinium 89																																															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3445UB0-1



Z22-3445UB0-1

WEDNESDAY, 15 JUNE 2022 – MORNING**APPLIED SCIENCE (Double Award)****UNIT 2: Space, Health and Life****HIGHER TIER**

1 hour 30 minutes

Section A**Section B****ADDITIONAL MATERIALS**

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5(b)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

You will need to refer to the separate Resource Folder to answer question **1**.

A Periodic Table is printed on page 16.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	25	
2.	7	
3.	14	
4.	10	
5.	9	
6.	10	
Total	75	

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Section A

Answer **all** questions in the spaces provided.

Refer to the separate Resource Folder to answer question 1.

1. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrtosiphon pisum*.

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	
Phylum:	
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	
Species:	
Scientific name:	<i>Acyrtosiphon pisum</i>

- (ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica*.



- I. State **one** advantage of using scientific names. [1]

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- II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

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(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

- (i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

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II. State **one** reason why this seasonal movement occurs. [1]

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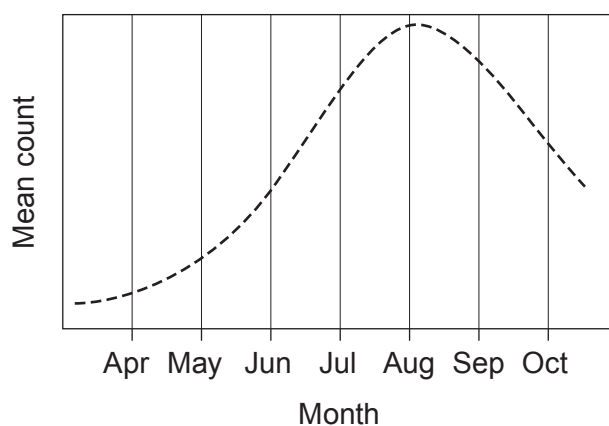
- (ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

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- (iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



- (iv) Use your line to explain the relationship between numbers of predators (frogs) and the numbers of their prey (Red Admirals). [3]

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- (c) (i) Use the information in **Table 2** to construct the part of the food web that depends on berries. [3]

Examiner
only



Berries



(ii) Explain why the frog can be found at more than one trophic level. [2]

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(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency =

8



Section B

Answer **all** questions in the spaces provided.

2. Medical imaging techniques include ultrasound, X-rays, CAT scans and MRI scans.

(a) (i) Describe how the two strands of DNA are joined in a double helix. [2]

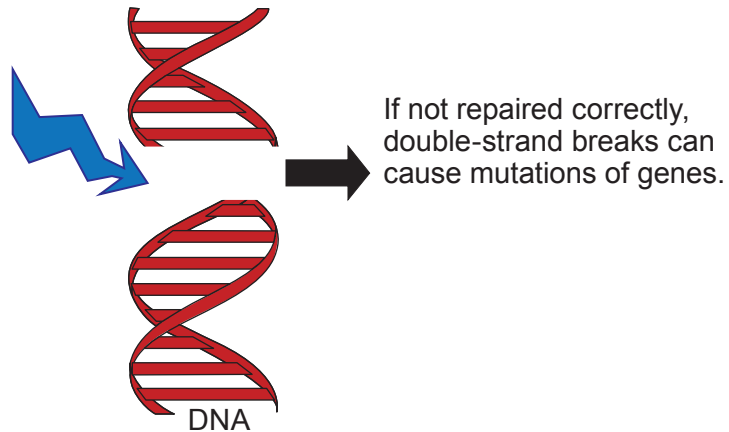
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(ii) Ionising radiation can have harmful effects on DNA within a human body. The diagram shows that it can cause a break in the double-strand.

Double-strand breaks produced
by ionising radiation, etc.



It is suggested that MRI scans are most likely to cause this kind of DNA damage and the other methods listed above are harmless. Explain whether you agree. [3]

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(b) Describe how an ultrasound image is formed. [2]

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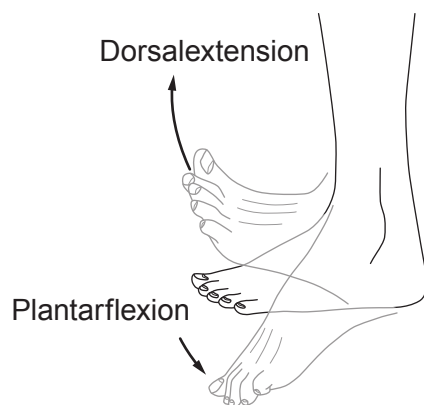
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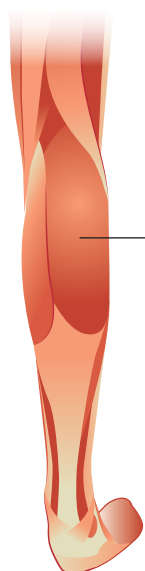
3. Healthy muscles and joints are essential for pain-free movement. These are very important for successful athletes.

- (a) The ankle is a type of hinge joint that is used when we walk and run. Its action is shown in the diagram below.



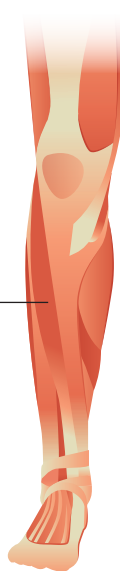
This movement is controlled by an antagonistic pair of muscles called the calf and tibialis anterior. The calf is at the back of the leg and the tibialis anterior is at the front of the leg.

Back of leg



Calf
muscle

Front of leg



Tibialis
anterior
muscle

Explain how this pair of muscles produce dorsalextension and plantarflexion.

[3]

.....

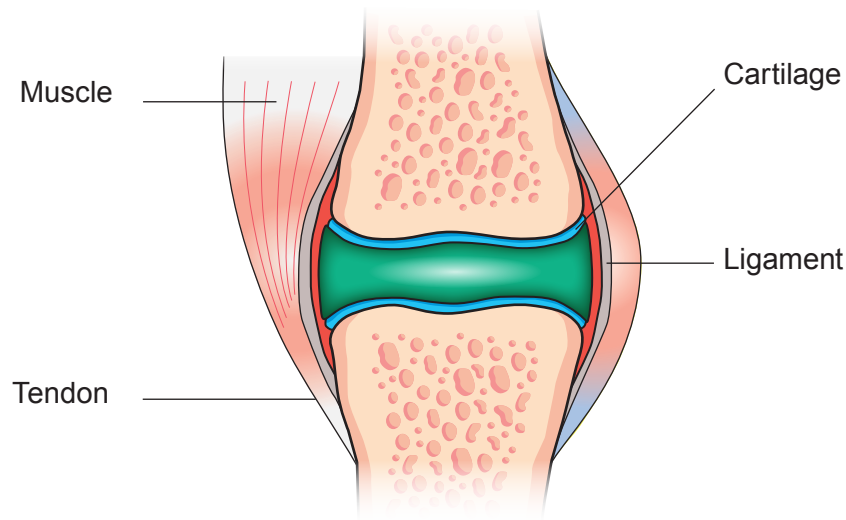
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- (b) The knee is a synovial hinge joint.



Explain how damage to cartilage and ligaments may affect an athlete.

[3]

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- (c) Two athletes, Dafydd and Tom, compete in a 400 m training race and their finish times are identical. Their times to travel certain distances within the race are measured and are shown in the table below.

Distance (m)	0	50	100	150	200	300	400
Tom's time (s)	0	7.65	13.65	20.48	27.30	41.93	56.55
Dafydd's time (s)	0	6.10	11.40	18.40	25.40	40.95	56.55

- (i) Dafydd thought his fastest speed was for the last 100 m of the race. Explain whether you agree.

[3]

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- (ii) Since they finished the race in the same time, Tom thought their maximum speed must have been the same. Plot graphs of distance against time for Tom and Dafydd and explain whether you agree with Tom. [5]



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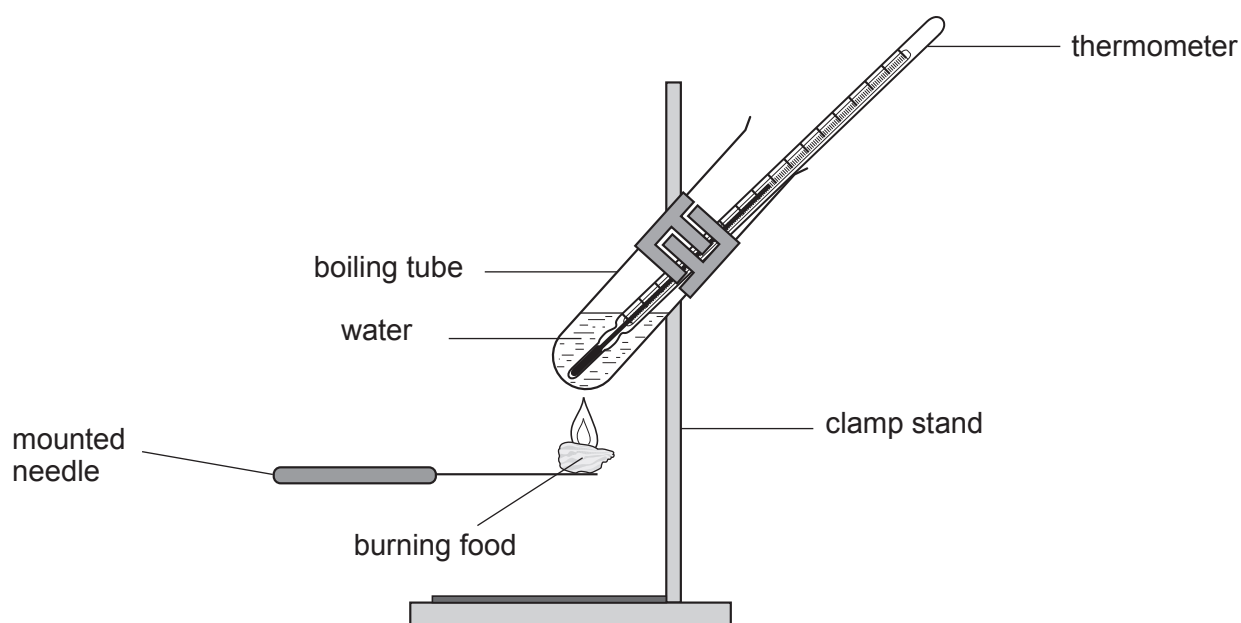
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4. The energy your body needs comes from the food you eat. The energy content of food can be measured in the laboratory.

- (a) The apparatus below can be used to measure the energy content of food in a school laboratory.



- (i) State **two** variables which should be controlled in this experiment **and** explain why they need to be controlled. [2]

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- (ii) Explain why the mean results obtained in the school laboratory are different from manufacturers' values given on a food label. [2]

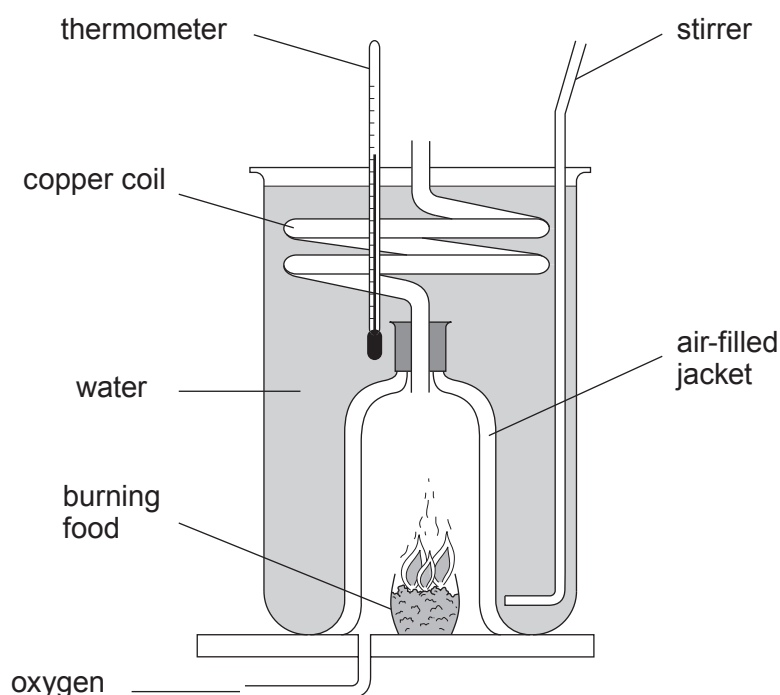
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- (iii) The energy content of food can also be measured using the calorimeter shown in the diagram below.



Explain why using this calorimeter is likely to improve the quality of the results. [3]

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- (b) The energy absorbed by a water sample is given by the equation

$$\text{energy (J)} = \text{mass of water (kg)} \times \text{temperature rise (}^{\circ}\text{C)} \times 4200$$

Results from the experiment for one food sample are shown below.

- mass of water = 20.0 g
- temperature rise = 15.4 $^{\circ}\text{C}$
- mass of food sample = 12.4 g

Use the information above to calculate the energy content of the food in J/g. [3]

$$1\text{g} = 10^{-3}\text{ kg}$$

energy content = J/g

10



6. Three images have been combined below to show colliding galaxy clusters 5 billion light years from Earth. The background is a Hubble Space Telescope image taken using visible light; the blue area is an X-ray image from the Chandra Space Observatory, and the red area is an image from a radio telescope.



- (a) (i) Compare the use of Earth-based and space-based telescopes to produce images of objects in space. [2]

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- (ii) Explain how the visible light received by the Hubble Space Telescope provides evidence for the origin of the Universe. [3]

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- (b) (i) State how long it takes radio waves to travel from the colliding galaxy clusters to reach the Earth. [1]

time =

- (ii) X-rays have a higher frequency than radio waves. Tick (✓) the box next to the correct statement below. [1]

X-rays take less time to travel the same distance as radio waves.

☐

X-rays take the same time to travel the same distance as radio waves.

☐

X-rays take more time to travel the same distance as radio waves.

☐

- (iii) The X-rays detected by the Chandra Observatory have a frequency range of 3×10^{16} Hz to 3×10^{19} Hz. They travel through space at a speed of 3×10^8 m/s.

Use the equation

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the maximum wavelength of these X-rays. [3]

maximum wavelength = m

☐

10

END OF PAPER





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1 4 He Helium 2																	
7 Li Lithium 3	9 Be Beryllium 4																
23 Na Sodium 11	24 Mg Magnesium 12																
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30						
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

16

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Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445UB0-1



TUESDAY, 13 JUNE 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life

HIGHER TIER

1 hour 30 minutes

Section A

Section B

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	8	
3.	6	
4.	7	
5.	10	
6.	9	
7.	8	
8.	16	
Total	75	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

- Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
- You may use a pencil for graphs and diagrams only.
- Write your name, centre number and candidate number in the spaces at the top of this page.
- Answer **all** questions.
- Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets at the end of each question or part-question.
- Question **5(a)** is a quality of extended response (QER) question where your writing skills will be assessed.
- You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
- You will need to refer to the separate Resource Folder to answer questions **1, 2 and 3**.
- A Periodic Table is printed on page 20.



JUN233445UB0101

Section A

Answer **all** questions.

Refer to the separate Resource Folder to answer the following questions.

1. (a) Use the information under **Figure 1** to answer the following question.
The wave speed of ultrasound in body tissue is 1540 m/s.

Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of the ultrasound used in a 2 MHz scan.

[3]

wavelength = m

- (b) Use the information in **Figure 4** to answer the following questions.

- (i) Select the most suitable radioisotope of **iodine** from the table in **Figure 4** that would be used as a tracer. Explain your choice.

[2]

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- (ii) The photograph shows a patient undergoing external beam radiation therapy for a brain tumour.



Select the most suitable radioisotope from the table in **Figure 4** that would be used for this treatment. Explain your choice.

[3]

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- (c) Refer to **Figures 4 and 6** in the Resource Folder to answer the following question.

State which of the radioisotopes **R1, R2, R3, R4, R5** or **R6**, is lutetium-177.
Explain your reasoning.

[3]

Radioisotope is lutetium-177.

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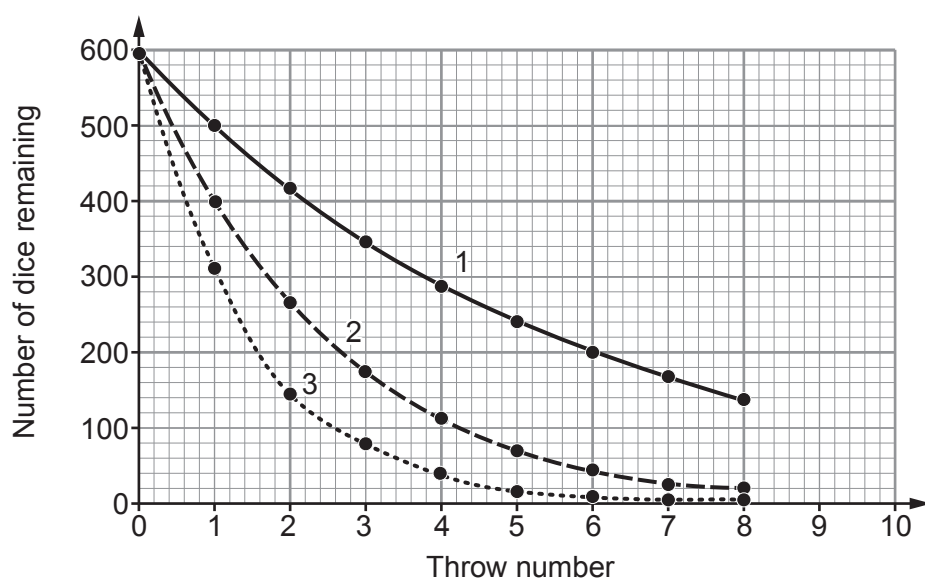
2. The following questions are based on the information on pages 6 and 7 of the Resource Folder about using dice to model half-life.

- (a) Each group's results were added together to give class results. Give **one** reason why this makes the data more repeatable. [1]

.....

.....

- (b) The results in **Figures 7 and 8** in the Resource Folder are used to plot graphs on the grid below.



- (i) Add lines to the graph to determine the half-life represented by line 3. [2]

half-life = throws



- (ii) Use the information in the graph to **tick (✓)** the boxes next to the **three** correct statements. [3]

Line 1 represents the data from Model 2.

☐

Line 2 shows decay of the quickest rate.

☐

Line 2 shows the equivalent of three half-lives after 5 throws.

☐

Line 1 represents the longest half-life.

☐

Line 3 would have been produced if 600 coins had been used instead of dice and the number of 'heads' counted after each throw.

☐

Line 3 has an activity that is $\frac{1}{8}$ of the original after 8 throws.

☐

- (c) The teacher suggests to the class that they add 5 red wooden cubes to the 50 dice they start with. The red cubes would not be removed after each throw but would be counted every time to improve the modelling of radioactive decay.

- (i) State what the red cubes represent. [1]

.....

- (ii) State how the presence of the red cubes would affect line 1 on the graph. [1]

.....



3. X-rays are commonly used to diagnose illness.(a) Refer to **Figure 2** to answer the following.

It is stated that X-rays have shorter wavelengths than ultraviolet rays but longer wavelengths than gamma rays. Explain whether you agree using data to support your answer. [3]

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(b) Refer to **Figure 3** to answer the following.

A patient has a series of X-rays taken following an accident. These were: one chest; one abdomen; and one skull X-ray. Calculate the percentage of the yearly background radiation the patient received from this combination of X-rays. [3]
(1 year = 365 days)

percentage =



Section BAnswer **all** questions.

4. Lichen can be used as an indicator species for air pollutants such as sulfur dioxide. Students performed an air pollution survey by recording the distribution of lichen around a city. Their results are shown in the table.

Distance from city centre (km)	Abundance of lichen (units)	Mean concentration of sulfur dioxide (units)
0.0	0	250
1.3	3	198
2.5	6	160
3.6	10	140
5.4	14	122

- (a) Use the information in the table to plot a graph showing the change in mean concentration of sulfur dioxide with the distance from the city centre on the grid opposite and draw a suitable line. [3]



Mean concentration of sulfur dioxide (units)

100
0

Distance from city centre (km)



- (b) Describe the relationship between distance from the city centre and mean concentration of sulfur dioxide. [2]

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- (c) Explain the effect of sulfur dioxide on the abundance of lichen. [2]

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5. (a) Explain why, for thousands of years, the percentage of carbon dioxide in the Earth's atmosphere remained approximately constant **and** how this stabilised conditions for life on Earth. [6 QER]

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- (b) During the past 200 years, evidence shows that the percentage of carbon dioxide in the atmosphere has been steadily rising.

One factor that contributes to this increase is how people travel to work.

The table shown below is taken from a climate change website.

It gives information about the mass of carbon dioxide produced **each journey per person** travelling to work by different methods each day.

Distance travelled (km)	10	20	25	50
carbon dioxide produced travelling by car (kg)	0.60	1.20	1.50	3.00
carbon dioxide produced travelling by bus (kg)	0.10	0.20	0.25	0.50
carbon dioxide produced travelling by train (kg)	0.18	0.36	0.45	0.90

Use this information to calculate the mass of carbon dioxide saved during a 5-day working week, if **250** people travel **30 km each way** to work and back home by train instead of every person using their own car.

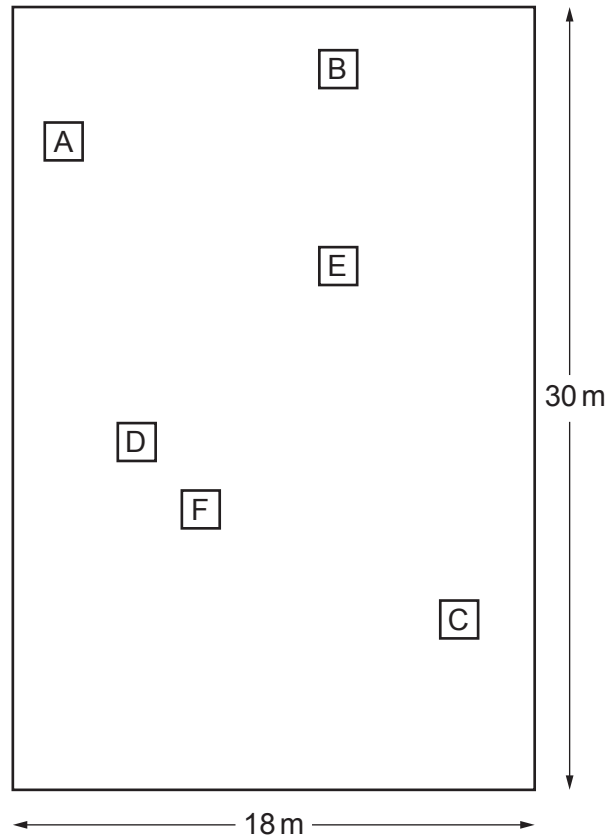
[4]

mass of carbon dioxide saved =kg

10



6. (a) Some students investigated the number of dandelion plants on a lawn. The diagram shows the lawn and the location of 6 quadrats (**A to F**) of area 1 m^2 which the students had placed at random on the lawn.



The students counted the number of dandelions in each quadrat and recorded their results in the table below.

Quadrat	Number of dandelions
A	6
B	3
C	2
D	7
E	4
F	5



- (i) Use the information to estimate the total number of dandelions on the lawn. [3]

estimated number of dandelions =

- (ii) The actual number of dandelions on the lawn is 1950.
Use the formula below to calculate the percentage error of your estimate above. [2]

$$\text{percentage error} = \frac{\text{estimated number of dandelions} - \text{actual number of dandelions}}{\text{actual number of dandelions}} \times 100$$

percentage error =

- (b) A species of dandelion has become resistant to one type of weedkiller. Explain, in terms of natural selection, how this has occurred. [4]

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7. Glucose homeostasis is an important biological process.

(a) Describe the role of insulin in controlling high blood glucose levels.

[4]

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(b) Describe the difference between type 1 and type 2 diabetes.

[2]

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(c) Describe the chemical test for glucose and state the colour change for a positive result.

[2]

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8. Cystic fibrosis is an example of an inherited disease caused by a recessive allele. A carrier of the disease has a genotype Ff.

- (a) Compare the chances of a child being born with cystic fibrosis if both parents are carriers, with a child whose father suffers from cystic fibrosis and a mother who is a carrier. [4]

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- (b) Cystic fibrosis causes a build-up of thick mucus in the lungs, which leads to poor gas exchange. Mucus in the lungs also allows increased bacterial growth that is responsible for infections.

Explain how the body defends itself against bacterial infection in the lungs. [3]

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- (c) Sufferers of cystic fibrosis benefit from being physically active to slow the rate of decline in lung function.

Explain the short-term and long-term effects of exercise on lung function. [3]

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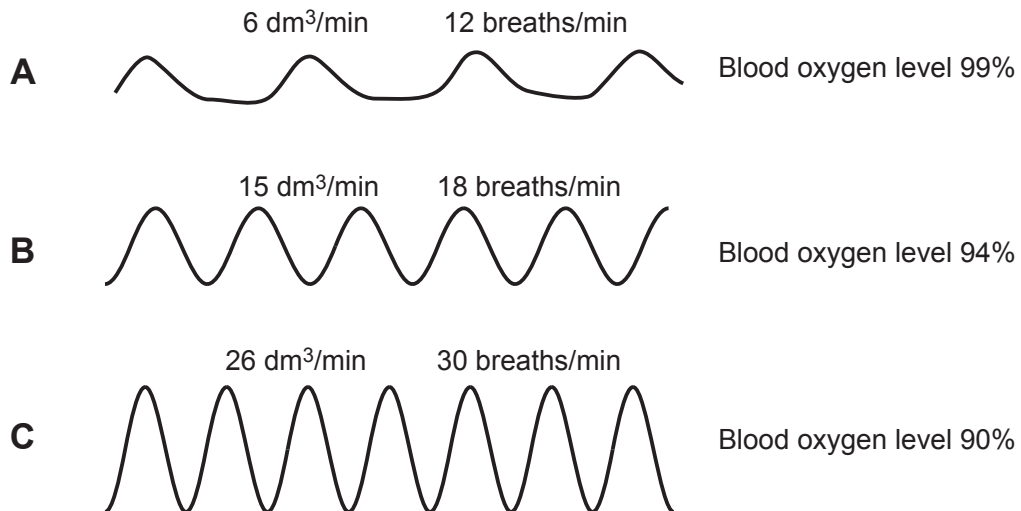


- (d) The diagram below gives information about the breathing of three different individuals, all of whom are resting.

A is a person who is not affected by cystic fibrosis.

B is a person with cystic fibrosis.

C is a person with cystic fibrosis and a lung infection.



The numbers show the volume of air breathed in per minute in dm³ per min and also the number of breaths per minute.

- (i) Use the equation:

$$\text{volume of air per breath} = \frac{\text{volume of air breathed in per minute}}{\text{number of breaths per minute}}$$

to compare the volume of air breathed in per breath for each of the breathing patterns.

[4]

A B C

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(ii) Compare the number of breaths per minute with blood oxygen levels in the three individuals. [2]

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Examiner
only

16

END OF PAPER



Group

0



6

7

0

1 H Hydrogen 1																			
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10		
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	
																		Key	

relative atomic mass

Symbol	Name	atomic number
A_r		Z